



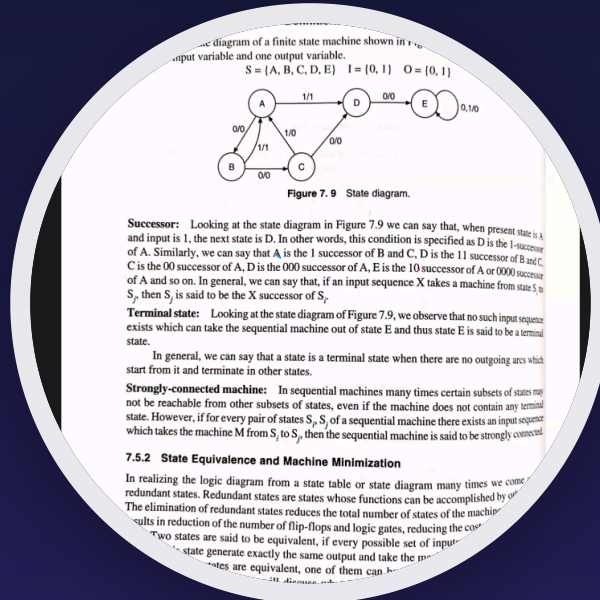
yuvin's Bedtime Adventure

A Magical Bedtime Adventure ✨



Bath Time

Once upon a dreamy night, little yuvin stepped into a magical
bedtime world filled with glowing stars, fluffy clouds, and beautiful
dreams ✨



Every sparkle in the sky whispered sweet bedtime stories while the
moon smiled softly above. Tonight was going to be a magical
adventure 🌙❤️



Pajama Time

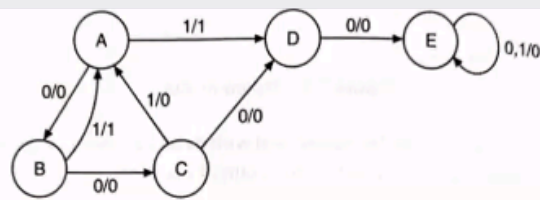


Figure 7.9 State diagram.

Successor: Looking at the state diagram in Figure 7.9 we can say that, when present state is A and input is 1, the next state is D. In other words, this condition is specified as D is the 1-successor of A. Similarly, we can say that A is the 1 successor of B and C, D is the 11 successor of B and C, C is the 00 successor of A, D is the 000 successor of A, E is the 10 successor of A or 0000 successor of A and so on. In general, we can say that, if an input sequence X takes a machine from state S_i to S_j , then S_j is said to be the X successor of S_i .

Terminal state: Looking at the state diagram of Figure 7.9, we observe that no such input sequence exists which can take the sequential machine out of state E and thus state E is said to be a terminal state.

In general, we can say that a state is a terminal state when there are no outgoing arcs which start from it and terminate in other states.

Strongly-connected machine: In sequential machines many times certain subsets of states may not be reachable from other subsets of states, even if the machine does not contain any terminal state. However, if for every pair of states S_i, S_j of a sequential machine there exists an input sequence which takes the machine M from S_i to S_j , then the sequential machine is said to be strongly connected.

7.5.2 State Equivalence and Machine Minimization

In realizing the logic diagram from a state table or state diagram many times we come across redundant states. Redundant states are states whose functions can be accomplished by other states. The elimination of redundant states reduces the total number of states of the machine which in turn

Now it's time for the softest, coziest pajamas! yuvin stretches little arms up high as sowmya helps with the buttons. The pajamas have the sweetest pattern, perfect for a night full of wonderful dreams.



Family Cuddle

7.5.3 Distinguishable States and Distinguishing Sequences

Two states S_A and S_B of a sequential machine are distinguishable, if and only if there exists at least one finite input sequence which when applied to the sequential machine causes different output sequences depending on whether S_A or S_B is the initial state. The sequence which distinguishes these states is called a distinguishing sequence of the pair (S_A, S_B) .

Consider states A and B in the state table shown in Table 7.2. When input X is 0, their outputs are 0 and 1 respectively and therefore, states A and B are called 1-distinguishable. Now consider states A and E. The output sequence is as follows:

$$X = 0 \quad \left\{ \begin{array}{l} A \rightarrow C, 0 \text{ and } E \rightarrow D, 0: \text{ outputs are the same} \\ C \rightarrow E, 0 \text{ and } D \rightarrow B, 1: \text{ outputs are different} \end{array} \right.$$

Here the outputs are different after 2-state transitions and hence states A and E are 2-distinguishable.

Table 7.2 Distinguishable states

PS	NS, Z	
	X = 0	X = 1
A	C, 0	F, 0
B	D, 1	F, 0
C	E, 0	B, 0
D	B, 1	E, 0
E	D, 0	B, 0
F	D, 1	B, 0

Again consider states A and C. The output sequence is as follows:

$$X = 0 \quad \left\{ \begin{array}{l} A \rightarrow C, 0 \text{ and } C \rightarrow E, 0: \text{ outputs are the same} \\ C \rightarrow E, 0 \text{ and } E \rightarrow D, 0: \text{ outputs are the same} \\ E \rightarrow D, 0 \text{ and } D \rightarrow B, 1: \text{ outputs are different} \end{array} \right.$$

The best part of the evening is cuddle time! yuvin
snuggles close with sowmya, feeling warm and safe.
Little arms wrap around for the biggest hug. Love fills
the room like a cozy blanket.



Story Time



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Here the

With a favorite book in hand, yuvin listens to magical tales of faraway lands. Each page brings new wonders and adventures. yuvin's eyes grow wide with amazement at every turn of the page.





Lullaby Time



Select one of the fol

☐ Chef

☒ Your name is: Ch

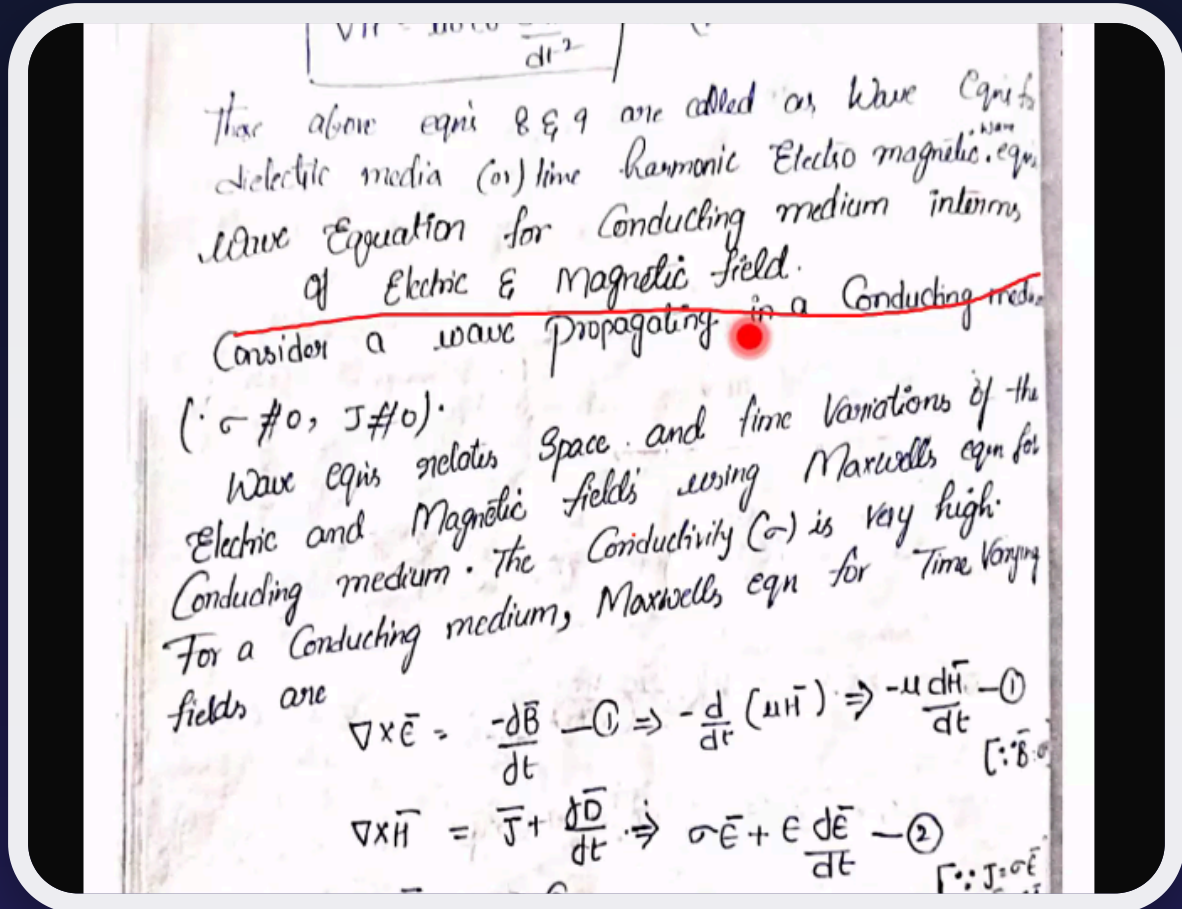
☐ Your name is:Che

A gentle song fills the air as sowmya hums a sweet lullaby. The melody dances through the room like fireflies in summer. yuvin's eyes begin to feel heavy, filled with peaceful dreams waiting to begin.





Goodnight Kiss

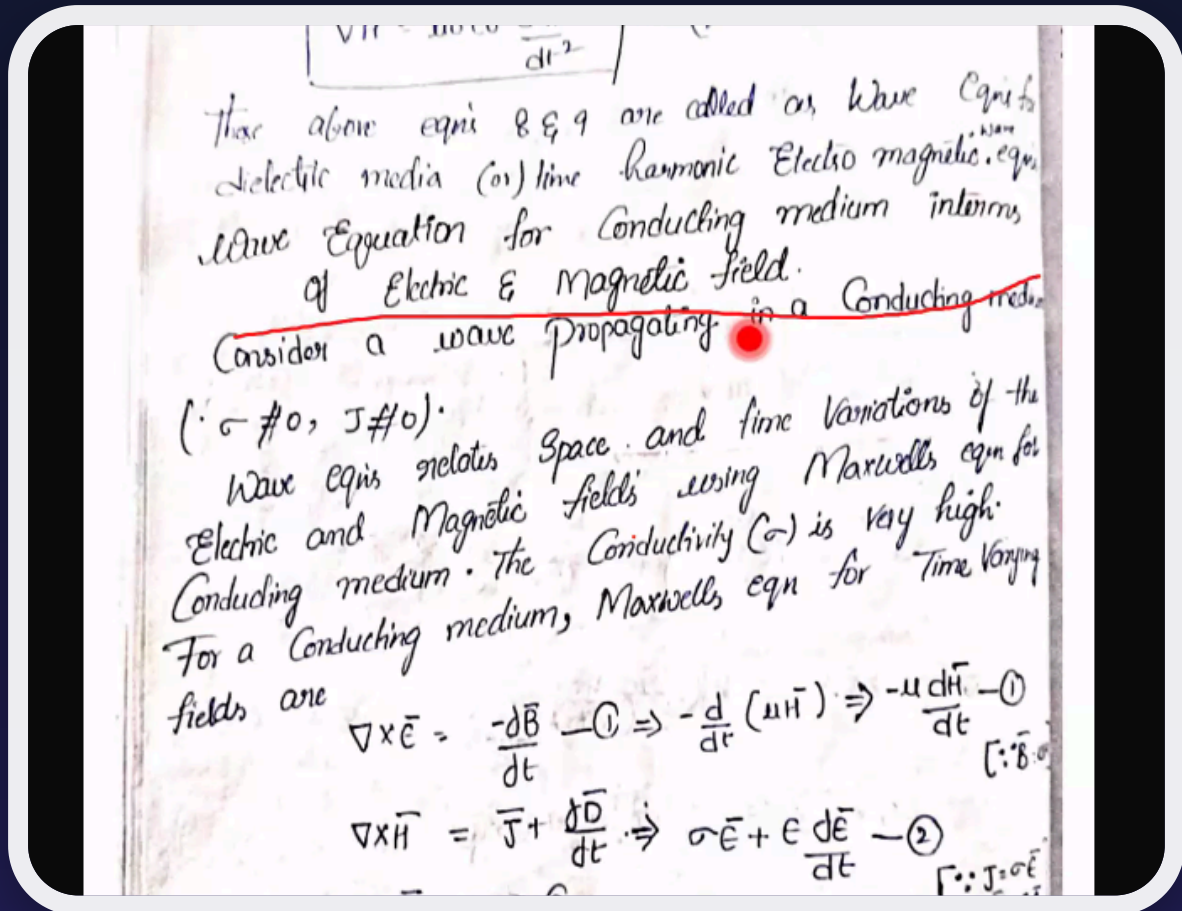


One sweet kiss on yuvin's forehead, then another on each cheek. 'Goodnight, little one,' whispers sowmya softly. The nightlight glows warmly, keeping watch through the night.





Sweet Dreams



And now, dear yuvin drifts off to dreamland, where unicorns play and stars sing lullabies. Tomorrow brings new adventures, but for now, sleep tight, little one.

Sweet dreams until morning light.

